

TKS Abstraction Ladder + Consciousness Unlock Spec (D: Novelty = A+B+C)

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This spec formalizes:

- 1) **How many abstraction layers** exist in the TKS agent pipeline
- 2) How **nesting depth p** adds “onion layers” of reconstruction/compression
- 3) A **Consciousness Unlock** mechanism where deeper recursion is **earned** by *validated novelty*:
 - (A) new TKS equation not seen before
 - (B) new RPM shortcut (fewer prerequisites / lower cost)
 - (C) new cross-domain association (FM■Male synthesis)

1) The Abstraction Ladder (fixed pipeline tiers)

These are the standard tiers your system moves through every episode.

L0 — Raw reality / raw input

- user text/audio
- tool outputs (web pages, files, logs)
- internal traces

L1 — Evidence objects

- raw_evidence_id pointers (hash, source tool, time, scope)

L2 — Extracted facts

- atomic statements pulled from evidence
- still human-readable, grounded by evidence IDs

L3 — RPM structure

- prerequisite cascade (tree)
- each node tagged: trigger, uncertainty, stakes, reversibility

L4 — TKS packets (symbolic summaries)

- equations/formulas that compactly encode the state:
- goal
- prereq state
- evidence links
- operators semantics

L5 — Canon-linked abstractions (semantic anchoring)

Each packet is mapped onto canonical dimensions:

- 4 Worlds
- 10 Noetics
- 40 Elements
- 7 Foundations
- 28 SubFoundations
- 22 Acquisitions
- Operators (+ − × ÷ ... as defined in your canon)

L6 — Nested memory nodes (precision p)

- hierarchical compression of packets using nesting
- deeper nesting = more compressed representation, reconstructable at chosen precision

L7 — Policy/templates (reusable patterns)

- reusable RPM motifs
- regulator timing heuristics (FM/ACBE/MVR)
- router preferences and tool strategies
- verifier threshold configurations

L8 — Meta-memory + governance history

- audit logs, replay sets, do-not-apply guards
- clearance tokens
- “why” traces for every decision

■ Fixed ladder size: **9 tiers (L0–L8)**

Then nesting depth adds **p** additional “onion” layers inside **L6**.

2) Nesting depth p (the onion layers)

Definition

- p = max reconstruction precision depth the system is allowed to expand
- p_max = current allowed depth (controlled by Consciousness Unlock)

Practical defaults

- Start: p_max = 2 (or 3 if you already have robust canon coverage)
- Hard cap early: p_max ≤ 5 until replay/eval is very mature

Reconstruction contract

- **p=0**: only the top packet (gist)
- **p=1**: gist + key prereqs summary
- **p=2**: adds evidence linkage + cross■references
- **p=3**: adds sub■prereqs + regulator activations
- **p=4–5**: deeper cross■world associations and multi■axis synthesis (advanced)

3) “Consciousness Unlock” (earned recursion)

Goal

Make deeper recursion / broader association a **hard■won prize**, not the default:

- stops runaway self■recursion
- rewards genuine insight
- creates developmental “stages”

What can be unlocked

An unlock may grant one or more of:

- 1) p_max += 1 (deeper nesting/reconstruction)
- 2) increase RPM depth cap (planning depth)

3) increase FM■Male association radius (broader retrieval)

4) increase rumination budget (offline iteration time)

4) What counts as Novelty (D = all of the above)

A candidate “novel idea” must be one of:

(A) New TKS equation

- new operator arrangement / new canonical mapping not previously stored
- not a trivial rewording of an existing packet

(B) RPM shortcut

- fewer prerequisites to reach the same success criteria, OR
- lower verified cost (tool_calls/latency/tokens), OR
- reduced uncertainty faster (ΔU improved)

(C) Cross-domain association (FM■Male synthesis)

- links two previously distant clusters in memory/canon
- produces a new, useful “superset” framing
- yields measurable improvement (reward/cost/uncertainty/coverage)

5) Validated Novelty (cannot be faked)

A novelty only “counts” if it passes **all** of:

1) **Validity (V)** — Verifier PASS (Module 2)

- canonical consistency
- evidence binding in High■Stakes/Critical

2) **Impact (I)** — measurable improvement (Module 4 + Module 5 replay)

- reward gain OR cost reduction without success loss
- uncertainty reduction

- coverage gain
- 3) **Newness ($1 - S$)** — low similarity to existing known templates/packets
- S is similarity (0..1). Lower is better.

Novelty Weight formula

$$NW = V \times I \times (1 - S)$$

Recommended thresholds:

- $NW \geq 0.70$ → “Heavy novelty” (instant unlock eligible)
- $NW \geq 0.45$ → “Counts” toward a token bank
- $NW < 0.45$ → logged as “interesting” but not counted

6) Computing the components (implementation guidance)

6.1 Validity (V)

V is derived from verifier outputs:

- $V = \text{confidence} \times \text{consistency} \times \text{evidence_ok}$

All normalized 0..1.

Hard rule:

- In High■Stakes/Critical, if evidence missing → $V = 0$ (HARD_FAIL)

6.2 Impact (I)

Impact is computed from replay■verified outcome deltas:

- reward gain (ΔReward)
- cost reduction (ΔCost)
- uncertainty reduction (ΔU)
- coverage increase ($\Delta\text{Coverage}$)

One simple normalized version:

- $I = \text{clip}_{01}(a\Delta\text{Reward} + b\Delta U + c\Delta\text{Coverage} + d\Delta\text{CostGain})$

6.3 Similarity (S)

Similarity is computed against stored packets/templates:

- vector embedding similarity (cosine) OR
- token/AST tree edit distance for TKS expressions

Recommended:

- Use **two** signals and take the max:
- $S = \max(S_{\text{embed}}, S_{\text{ast}})$

7) Depth Permission System (DPS)

7.1 State

```
{
  "p_max": 2,
  "novelty_tokens": 0,
  "cooldown_until_episode": "ep_...",
  "last_unlock": "ep_...",
  "risk_bias": {
    "high_stakes_unlocks": "disabled",
    "critical_unlocks": "disabled"
  }
}
```

7.2 Token rule (quantity gate)

- If a novelty has **NW** ≥ 0.45 and passes replay:
- `novelty_tokens += 1`
- If `novelty_tokens` $\geq N$ then:
- `p_max += 1`
- `novelty_tokens = 0`

Recommended starter:

- `N = 5`

7.3 Heavy novelty rule (weight gate)

If a novelty has **NW** ≥ 0.70 and passes replay:

- unlock one of:

- $p_max \ += \ 1$ OR
- one-time “depth burst” (temporarily allow p_max+1 for a rumination cycle)

7.4 Cooldown (anti-mania)

- After an unlock: set cooldown for K episodes (e.g., $K=10$)
- During cooldown: token accumulation allowed, but no unlock

7.5 Failure penalties

If a supposed novelty causes:

- verifier `HARD_FAIL`
- policy violation
- regression on replay

Then:

- subtract tokens (e.g., `novelty_tokens -= 1` with floor at 0)
- optionally reduce p_max by 1 if repeated

8) Safety and governance constraints

8.1 High-Stakes / Critical constraints

- **Critical:** no unlocks. No deeper recursion. Only ASK/SLOW/VERIFY.
- **High■Stakes:** unlocks disabled by default.
- Enable only after stable eval sets and admin approval.

8.2 Uncertainty-drop rule (anchoring)

If deeper recursion is used and uncertainty does not drop after 2 attempts:

- stop deepening
- return to ASK for missing constraints/evidence

8.3 Hard caps (early deployment)

- RPM depth cap: 3 (Normal), 2 (High■Stakes initial)
- Tool call cap: 12 (Normal), 8 (High■Stakes), 0 (Critical until verified)

- p_max hard cap early: 5

9) Events and logging (required for replay)

Every novelty candidate produces a log event:

9.1 NoveltyCandidate event

```
{
  "event_type": "NOVELTY_CANDIDATE",
  "episode_id": "ep...",
  "novelty_type": ["A", "B", "C"],
  "tkn_packet": "...",
  "rpm_delta": {"prereqs_before": 7, "prereqs_after": 5, "cost_gain": 0.12},
  "association": {"cluster_a": "...", "cluster_b": "...", "bridge_packet": "..."},
  "scores": {"V": 0.81, "I": 0.62, "S": 0.22, "NW": 0.392},
  "verifier": {"status": "PASS", "gate": "ALLOW"},
  "replay": {"passed": true, "reward_delta": 0.06}
}
```

9.2 DepthUnlock event

```
{
  "event_type": "DEPTH_UNLOCK",
  "episode_id": "ep...",
  "old_p_max": 2,
  "new_p_max": 3,
  "reason": "tokens >= 5 OR heavy_novelty",
  "policy_versions": {"router": "r1.3", "verifier": "v1.1"}
}
```

10) Practical starting configuration (recommended)

- Start p_max=2
- Tokens threshold N=5
- Heavy novelty threshold NW>=0.70
- Count threshold NW>=0.45
- Cooldown K=10 episodes
- Unlocks disabled in High Stakes/Critical (until your eval suite is mature)

- Cross-domain association allowed to expand **only** when verifier PASS and canon coverage is sufficient

11) Canonical TKS expression hooks (optional, but useful)

You can represent the unlock mechanism inside TKS as a meta-layer, e.g.:

- $NW = V \times I \times (1 - S)$
- $UNLOCK(p) = (NW \geq \theta w) + (Tokens \geq N) - (Violation)$

But keep these as **policy/meta** expressions rather than core canon until you intentionally add them.

End of Spec.